

Article

A Study of the Dynamic Evolution Game of Cooperative Management by Multiple Subjects Under the Forest Ticket System

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Abstract: As a kind of proof of the rights and interests of forest resources, the implementation of forest ticket is an important measure for China to revitalise collective forest land resources and promote the transformation of forest resources into economic development advantages, which is of great significance to the sustainable development of forestry. Based on the dynamic evolutionary game method of multiple subjects, this paper constructs an evolutionary game model of state-owned forest farms, village collective economic organisations, and forest farmers; analyses their strategy choices and the stability of the equilibrium point of the game system; and examines the influence of different parameter values on the model strategy evolution through combination with numerical simulation methods. The results show that the level of knowledge and participation in the forest ticket system significantly influence the optimal equilibrium strategy of each subject of forest cooperative management. The optimal strategy is only when the evolution of the game model is stable at a high level of input, participation, and high willingness to participate. In addition, the forest market environment and the level of inputs from the agents also affect the rate of stabilisation of behavioural strategies.



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Keywords: forest ticket system; cooperative forest land management; multiple subjects; evolutionary game

1. Introduction

Forests are an important component of the Earth's ecosystem, providing diverse ecological services and rich material goods for the development of human society [1], and playing an irreplaceable role in maintaining ecological balance and mitigating climate change. However, with the advancement of urbanisation and modernisation in developing countries, the expansion of town area has accelerated, and the forest area has been continuously squeezed by urban space [2]. The over-exploitation and over-use of forest resources as a result of rapid industrialisation have led to the depletion of forest resources far in excess of their growth [3]. Forests in some developing countries are being rapidly destroyed, with adverse effects such as increased soil erosion, loss of biodiversity, and increased greenhouse gas emissions that contribute to climate change [4,5]. At the same time, the development of non-farm jobs has led to a decrease in the proportion of forestry income in household income, and the dependence of farm households on forestry has become smaller. Forestry is no longer the main livelihood industry of farm households, forestry professional management is hindered, and the phenomenon of forest land desertion is constantly apparent [6,7]. The Global Forest Resources Assessment 2020 of the Food and

Agriculture Organization of the United Nations (FAO) shows that the global forest area continues to shrink, and the ecological and economic functions of forests are continuously damaged [8]. The situation is particularly serious in areas where developing countries are concentrated, such as Asia, Africa, and Latin America. This poses an existential threat to forest communities that depend on forest resources for their livelihoods and greatly limits socio-economic and ecological development [9]. The United Nations Forest Instrument (UNFI) emphasises the multiple economic, social, and environmental benefits of forests and trees, pointing out the important contribution of sustainable forest management to the sustainable development of forestry and poverty eradication [10,11]. Strengthening the political commitment to practicing sustainable forest management and developing and implementing national strategies and plans for sustainable forestry development have become major concerns for national leaders and policy makers.

In China, according to the *Forest Law of the People's Republic of China*, forest resources are divided into state-owned forests and collective forests, based on their ownership. As the country with the largest area of planted forests in the world, China's collective forest areas, which cover more than 60% of the country's forest area, overlap significantly with economically underdeveloped regions, affecting the livelihoods and well-being of more than 500 million farmers [12]. However, for a long time, China's collective forests have faced problems such as small and scattered forest land management areas, insufficient vigour, low efficiency, and uneconomical scale [13,14]. These problems have led to the low productivity level of collective forests, weak supply capacity of ecological products, and insufficient contribution to farmers' income [15]. At the same time, as China's urbanisation process continues to advance, the rural labour force continues to move out of the rural areas [16]. The intensification of rural hollowing out has increased the opportunity cost of collective forest management, which is not conducive to the development of rural economy [17]. In order to improve the efficiency of collective forest management, forestry co-operative management and large-scale management have become important choices for the development of collective forest management [18,19]. In this context, how can forest management strategies transform forest resources into economic development advantages, both to enhance the efficiency of sustainable forestry management and to meet the needs of socio-economic development. This has become an urgent need for China to promote the sustainable development of forests and participate in the realisation of the global sustainable development goals.

As the world's largest developing country, China, in order to revitalise its forest resources and unleash the potential of forestry management, has adopted a series of measures, such as the reform of collective forest rights, which have provided a path for new explorations in realising the transformation of forestry ecological resources into economic capital [20–23]. As a pioneer in forest management system reform, Sanming City, Fujian Province, has boldly innovated its forest management policy by introducing a forest ticket system [24]. The forest ticket system is essentially an innovative model of the cooperative management of forest land. Specifically, the forest ticket system is promoted by government departments and is based on joint operation, involving multiple subjects. State-owned forest farms, village collective economic organisations, and individual forest farmers are all major participants in the operation [25]. In the operation of the forest ticket system, the state-owned forest farms, as the leading operators of forest land, invest in capital, technology, and management, dominating the whole process of operation and management in an absolutely controlling position. The village collective economic organisation, as a grassroots economic community, invests in forest land and its management rights in the name of the collective by means of proxy. Forest farmers, as free individuals, in full consideration of their own will, through the form of cooperation or non-cooperation with

the village collective economic organisation, will transfer the management rights of forest land to the state-owned forest farms in the form of investment in forest land and shares.

As an important carrier of this system, the forest ticket plays an important role as a certificate of the rights and interests of each participating subject in the cooperative management of forest land. It is issued on the basis of the amount of value of forest rights under the cooperative framework, which is prepared and issued by state-owned forestry enterprises and institutions according to the share of investment taken by village collectives or individuals. At this point, the forest stamps owned by the collective and forest farmers have the characteristics of being transferable and pledgeable. Finally, under the state of a complete transaction system, according to the individual's wish, the foresters can, according to the process of filing the transaction record of forest stamps, list the forest stamps in the property rights centre for trading or apply for a pledge loan from financial institutions. In the case of forest ticket holders wanting to withdraw from the forest ticket business, the state-owned forestry enterprises and institutions are obliged to provide buyback services to protect the rights and interests of forest farmers and the protection of forest resources during the transaction process of forest tickets. The government, as the author of the forest ticket system, does not directly intervene in specific business activities during the whole process of circulation of forest tickets, but is only responsible for guidance and supervision. Mechanism of the operation of the Sanming Forestry Certificate practice is shown in Figure 1. At present, the forest ticket system is mainly implemented for newly planted forests, near-mature forests and mature or over-mature forests in artificial commercial forests (this information is provided by the Forestry and Grassland Bureau of Sanming City, Fujian Province at https://www.sm.gov.cn/zw/zwx/sjdt/202204/t20220421_1778897.htm, accessed on 12 March 2025), which ensures the effective implementation of the forest ticket system and protects the rights and interests of forest farmers. In this way, the forest ticket promotes the rational use of forestry resources and provides a stable source of income for forest farmers.

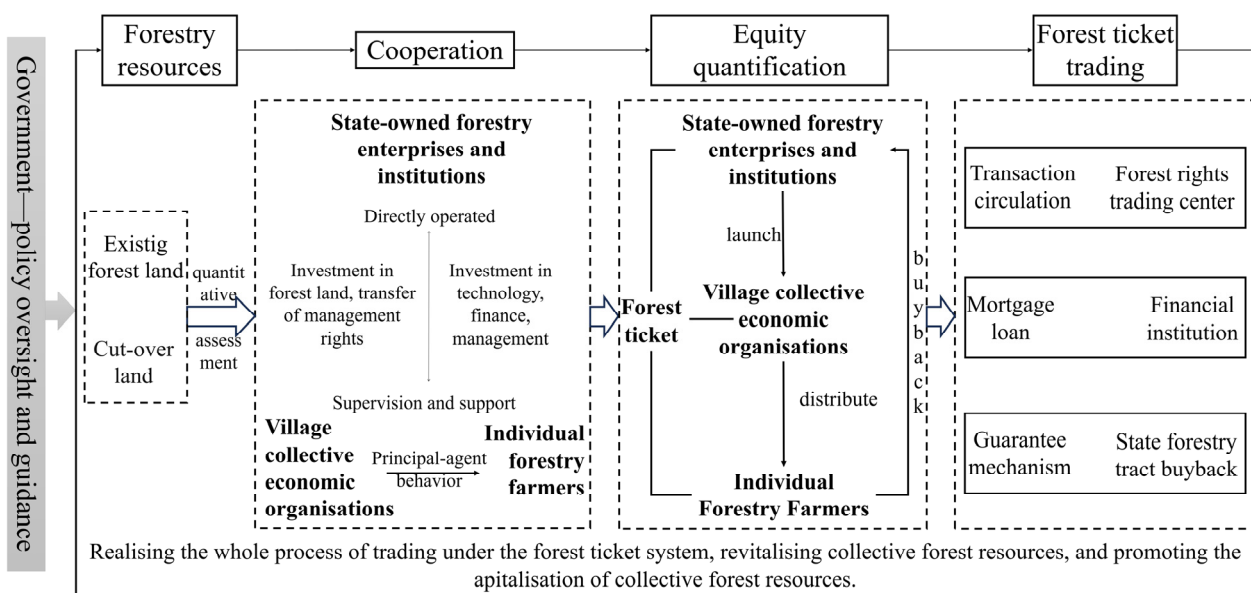


Figure 1. Mechanism of operation of the Sanming forest ticket practice.

The original design of the forest ticket system was to promote large-scale and intensive forestry operations and to transform the forest ecological resources into economic value through the transfer of forest management rights and the cooperative management of forest land, while increasing the income of forest farmers. However, in the current pilot operation process, there are still problems that need to be solved in the cooperative management of

forest land. As the main participants of the forest ticket system, the behavioural choices of state-owned forest farms, village collective economic organisations, and individual forest farmers affect the results of the forest ticket system to a certain extent. State-owned forest farms are limited by the long forestry production cycle, capital investment needs, and other issues; in the forest ticket system of forest land management, the process of financing, and loan repayment pressure, effectively achieving the economic and ecological benefits of growth is difficult in the short term [26]. Village collective economic organisations and individual forest farmers are limited by the low level of awareness of the policy, low income from forest tickets, unbalanced benefit distribution mechanism, imperfect risk management system and transaction mechanism, etc., and there is a certain degree of resistance to the forest ticket system [27–29]. Comprehensively speaking, the current forest land cooperative management is facing the problem that the government departments are highly motivated to promote it, but the participation of other stakeholder subjects is insufficient. In addition, the rotation period of forest land trees is long. During one or even more rotations, the interests of each subject are easily affected by intergenerational relationships and other influences that make it difficult to maintain a balance [30]. This has led to the lack of stability in the cooperative relationship between the subjects of forest land cooperative management, the difficulty of promoting the forest ticket system, and the willingness to cooperate needs to be improved. Therefore, under the background of vigorous promotion by government policy, how can the participation of all subjects of forest land management in forest land cooperation be promoted and the motivation of forest land cooperation participation be stimulated? How can the interests of each subject in the cooperative forest management be balanced? How can a long-term sustainable cooperative relationship of the forest ticket system be built through the perspective of the interest game of each subject? These questions have not been fully explained and answered.

Existing research on the forest ticket system mainly focuses on discussing the co-operation mode and implementation status of the system from the perspectives of the transfer of forest land management rights and the subject of cooperation [31–33], and the four-in-one cooperative management mode has been proposed [34]. However, few studies have conducted in-depth discussions of the insufficient participation of various subjects in the cooperative management of forest tickets and the difficulty in realising their interests, as well as insufficient analyses of the intrinsic interest mechanism of cooperative decision-making on forest land. In addition, China shares many characteristics of forestry development and rural economy with other developing economies. How to solve the problem of economic development and forestry protection and how to turn forest resources into economic development advantages are common concerns of developing countries. In view of this, this paper takes the forest reform system in Sanming City, Fujian Province, China, as an example based on the perspective of the synergy of multiple subjects, focuses on the core interests of the forest ticket system, explores the optimal equilibrium strategy of the three cooperative management with the help of the evolutionary game analysis tool, and proposes an optimisation path for the development of the forest ticket system based on the results of the numerical simulation.

The possible marginal contributions of this paper lie in the following points: (1) Through the multi-subject evolutionary game and case analysis, it not only identifies the respective interests of the participating subjects of the forest ticket system but also provides a theoretical basis for the choice of decision-making paths on how to promote the long-term effective operation of the forest ticket system. (2) The analysis of the cooperation mechanism from the perspective of the interest game of multiple subjects fills the gap in the study of the forest ticket system. (3) It provides guidance for solving problems such as forest ecological damage brought by urbanisation and modernisation development,

and promoting policy innovation for sustainable forestry operation, which has important theoretical value and practical significance. (4) In addition, taking the world's largest developing country as an example, the research in this paper can provide theoretical support and strategic references for developing countries to improve their sustainable forest management strategies, establish effective input and benefit distribution mechanisms, and further contribute to the realisation of global sustainable development goals.

2. Theoretical Analysis and Model Setting

2.1. Theoretical Foundation

2.1.1. Stakeholder Theory

Stakeholder theory originated in business management and has since been widely used in a variety of fields, including investment, macro management, and public governance [35]. The theory, first proposed by Freeman, emphasises that individuals and groups that have a direct or indirect impact on the achievement of organisational goals should be taken into account in decision-making, and that the outcome of decision-making is not the maximisation of owners' interests [36]. In the process of achieving organisational goals, stakeholder groups have invested in production factors, shared organisational risks, and supervised the decision implementation process. Therefore, decision-making should fully consider the demands of stakeholder groups so as to reach the optimal strategy of multi-level, grid, and strong associations. The forest ticket system requires the participation of multiple subjects, such as state-owned forestry enterprises and institutions, village collective economic organisations, and individual forest farmers. It is a system composed of multiple stakeholder decision-making subjects with heterogeneity and expected interactions. The state-owned forest farm is the initiator of the forest ticket system, which holds 51% as the main operating body and has greater dominant power over the operation of the forest ticket system. In the acquisition of social information and understanding of the system, although there is a certain weakness in the village collectives and forest farmers, the two are more familiar with the village, people, and forest conditions. They are the key subjects indispensable to the long-term, sound operation of the forest ticket system. Under the forest ticket system, the three parties form a stakeholder group. But, due to the different characteristics and roles of the three parties, the relationship of interests is more complex, so that the implementation of the forest ticket system cannot be achieved in one go. In this regard, this study is based on the stakeholder theory, taking into full consideration the demands of different stakeholders, and discusses the cooperative management of the forest ticket system.

2.1.2. Evolutionary Game Theory

Evolutionary game theory breaks through the assumption of 'complete rationality' of traditional game theory and solves the problems caused by incomplete information, limited cognition, and biased judgement in reality. Different from traditional game theory, evolutionary game theory originates from Darwin's idea of biological evolution, which is a theory that combines the analysis of game theory and the analysis of dynamic evolutionary process [37]. Evolutionary game theory is based on the finite rationality hypothesis [38,39]. The theory holds that under the condition of incomplete information, limited rational individuals or groups, according to their vested interests, constantly perform trial and error, adjust, learn, and improve their own behavioural decisions; adjust their strategies at the margin to pursue the improvement of their own interests; and ultimately reach the optimal behavioural strategy [40,41]. In recent years, with the continuous deepening of research, evolutionary game theory has continued to deepen in the field of complex system governance. Existing studies generally adopt multi-subject evolutionary game

models, combined with a numerical simulation and stability analysis, to reveal the strategy interaction law of limited rational subjects through a dynamic evolutionary mechanism. This has achieved a wide range of applications in the directions of artificial intelligence [42], ecological governance [43], policy tool synergy [44], corporate economics [45], and cross-domain cooperation [46].

At present, the forest ticket system is still in the exploratory stage; the cognitive level, decision-making ability, and information acquisition ability of all parties are often one-sided; and the behavioural logic is driven by interests with different tendencies. Therefore, participants need to make trial and error in the process of the limited rationality game and optimise the adjustment to seek better strategic choices. In addition, the cooperative management of the forest ticket system is a long-term and dynamic process, and the evolutionary game theory can fully take into account this dynamism, reflect the behavioural strategies and evolutionary trajectories among the subjects of the cooperative management of the forest ticket in a more objective manner, and present the reality of the development situation. Based on this, this study adopts the evolutionary game model to analyse the equilibrium point of interests among the subjects of cooperative management under the policy background of the forest ticket trading system, and to clarify the stabilisation strategy of the cooperative management of the forest ticket through the participation of multiple subjects.

2.2. Analytical Framework

Under the forest ticket system, the government, state-owned forest farms, village collective economic organisations, and individual forest farmers each play important roles. The government, as the leading force of policy guidance, promotes the in-depth reform and innovation of the collective forest right system, mainly through the formulation of policy measures and strategic leadership, to stimulate the state-owned forest farms, village collective economic organisations, and individual forest farmers to participate in cooperation and regulates the cooperative management, but does not directly intervene in the details of the operation of the forest ticket practice [23,47]. Therefore, based on the specific operation of forest land cooperation under the forest ticket system, this study mainly constructs a tripartite main body game research framework of ‘state-owned forest farms–village collective economic organisations–individual forest farmers’, as shown in Figure 2.

As an important leader of forest land management in the forest ticket system, state-owned forest farms shoulder the important responsibilities of forest resource management and protection, as well as promoting the sustainable management of forest land resources [48,49]. In the process of cooperation with village collective economic organisations and individual forest farmers, state-owned forest farms need to coordinate the interests of all parties in order to achieve a win–win situation [50]. By investing in forest resources, village collective economic organisations and individual forest farmers can support the ecological and economic goals of the state-owned forest farms and receive corresponding benefits from them. However, the willingness of these two to participate in co-operation is constrained by the guarantee of benefits, which mainly stems from the influence of trust risk and information asymmetry. Under the forest ticket system, village collective economic organisations act as a bridge and intermediary, establishing an organisational mobilisation and principal-agent relationship with individual foresters [20,51]. In this relationship, the two share common interests in terms of poverty alleviation, income generation, and strengthening social ties, but they also face the problem of the unequal distribution of benefits, especially in areas where clan concepts are stronger, which may further impede the smooth advancement of cooperation [13,52]. Individual forest farmers, as owners of forest land management rights and forest tree use rights, can participate in

forest land cooperation under the forest ticket system by means of forest land shareholding so as to achieve the unbundling of human–land relations and income enhancement. However, as individual foresters face the dilemma of limited rationality and information asymmetry, this often leads to hesitation in the decision-making process [12,53]. For state-owned forest farms, the unpredictability of individual farmers' behaviours must also be taken into account, and the resulting risks must be borne.

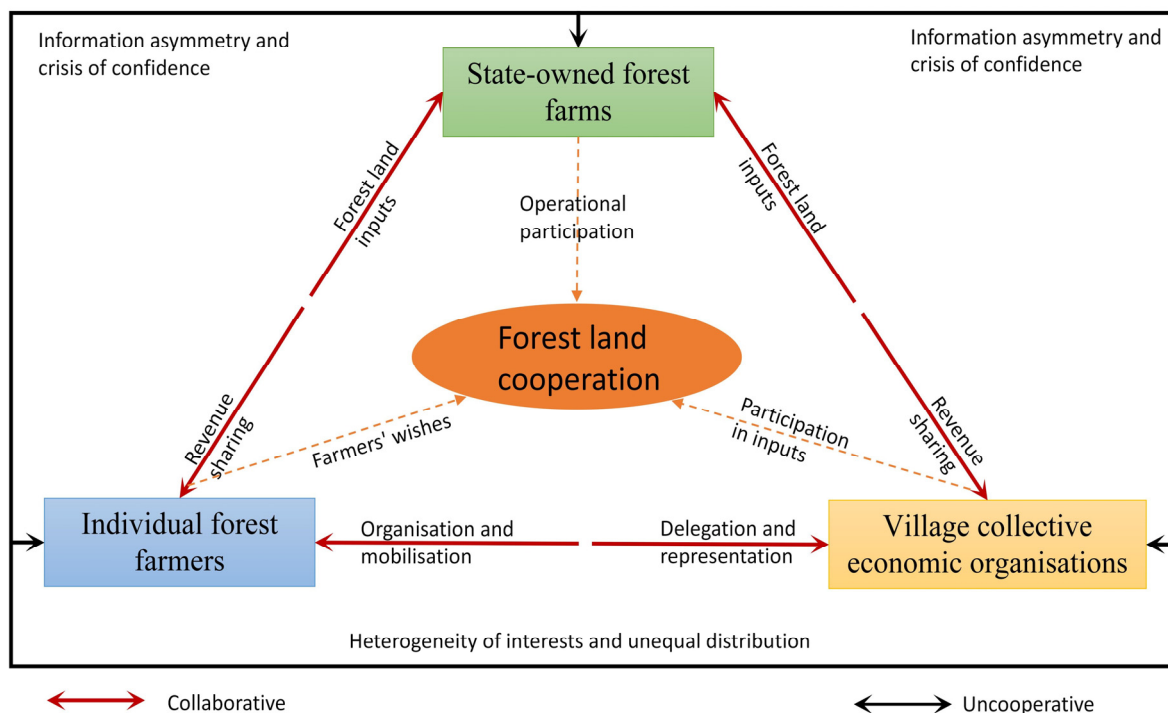


Figure 2. Multi-subject relationship of forest land cooperation under the forest ticket system.

2.3. Research Area and Model Setting

2.3.1. Selection of the Research Area

Sanming City, Fujian Province, is rich in forest resources, with a forest coverage rate of 78.73% and a high proportion of collective forests. Vegetation cover is shown in Figure 3. It is a key collective forest area in southern China, a national pilot area for collective forest reform, and an important forest product processing base in Fujian Province. As the origin of China's collective forest rights system reform, Sanming City has actively explored the reform of the forest ticket system. In accordance with the Law of the People's Republic of China on Rural Land Contracting and related policies, Sanming City has formulated the Sanming City Forestry Ticket Management Measures in light of the need to deepen the reform of the collective forest rights system, which are used to guide government departments at all levels to effectively promote the construction of the forest ticket system. In 2019, Shaxian District of Sanming City, in response to the new era of rural revitalisation strategy and the requirements of state-owned forest farm reform, took the lead in the province to launch the 'forest ticket system' reform [54] and issued the first batch of forest tickets in Gaosha Township on 23 November. This provides practical support for the discussion in this study.

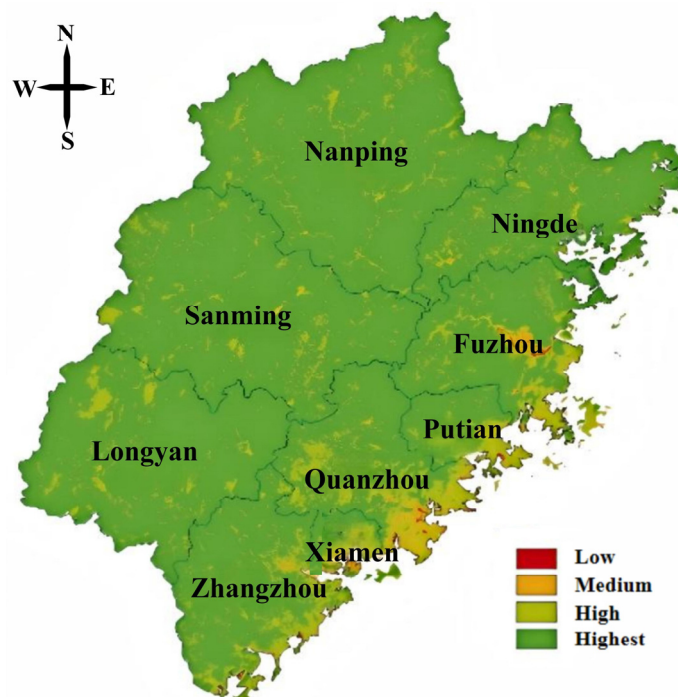


Figure 3. Vegetation cover in Fujian Province.

2.3.2. Model Assumption

Combined with the practice case and management approach of forest tickets in Sanming City, in the cooperative forest management model under the forest ticket system, the state-owned forest farms, village collective economic organisations, and individual forest farmers, as the direct stakeholders of the cooperative management, are established as the main bodies of the tripartite evolution game model, and the following model assumptions are proposed. The relevant parameters are designed as shown in Table 1.

Table 1. Parameter settings.

Parameter	Implication
R	Revenue from the cooperative management of forest land under the forest ticket system
a	Proportion of the value of forest tickets allocated to state-owned forest farms
b	Proportion of the value of forest tickets allocated to village collective economic organisations
d	Proportion of value of forest tickets distributed to individual foresters
θ	Forest business environment: expressed by timber price index. The market price of timber in the year before the implementation of the forest ticket system is taken as the base year, >1 indicates that the forest management environment is favourable, while <1 indicates that the forest management environment is deteriorating.
g	Probability of state-owned forest farms choosing {high level of input} strategy
c	Probability of village collective economic organisation choosing {participation} strategy
s	Probability that individual forest farmers choose {high willingness to participate} strategy
C_1	Costs of the high-level input operation of state-owned forest farms, including inputs such as manpower, management costs, etc.
B	Policy implementation benefits obtained by state-owned forest farms (proceeds from government subsidies and policy incentives received by state-owned forest farms after their participation in forest ticket cooperation)
p	Degree of low-level input in state-owned forest farms (quantifies the extent of low levels of inputs by analysing the difference in input ratios from the base year through financial data)
L_1	Reputation loss due to a failure to maintain good relations with surrounding communities and farmers when state-owned forest farms have low levels of inputs (assessment of the extent of the decline in the reputation of state forest farms through community satisfaction surveys and correlation analyses with the loss of revenue from low inputs)

Table 1. Cont.

Parameter	Implication
L_3	Loss of individual forest farmers with high willingness to participate when state-owned forest farms have low levels of inputs (difference between foresters' income from forest tickets and the base year when there was a low level of inputs from state-owned forests)
C_2	Costs of village collective economic organisations' participation in cooperation, including inputs such as publicity and popularisation, cooperation negotiations, and meeting organisation
C_4	Additional costs for individual forest farmers with high willingness to participate when village collective economic organisations do not participate in cooperation (analyses the financial cost of additional inputs for foresters through questionnaires and interviews with relevant data)
L_2	Loss of weakened social ties with individual forest farmers when village collective economic organisations do not participate in the cooperation. (analysis of changes in foresters' social trust, willingness to cooperate, and social support when village collective economic organisations are not involved through questionnaires and interviews, and an analysis of correlations with economic losses)
M	Forest land use fees paid by the state-owned forest farms to village collective economic organisations during the main harvesting of forest trees
C_3	Costs when individual forest farmers' willingness to participate is high, including inputs such as information collection and institutional learning
n	Degree of inputs when individual forest farmers' willingness to participate is low (quantification of the extent of inputs in the case of low willingness through the difference between the cost inputs as a proportion of household expenditures and the base year)
G	Default risk of state-owned forest farms with high level of inputs when individual forest farmers' willingness to participate is low (correlation analysis based on foresters' willingness to participate and frequency of defaults through questionnaires and interviews)
m	Probability of individual forest farmers' participation in forest ticket trading
A	Benefits gained by individual forest farmers from forest ticket trading
h	Proportion of prestige gains through individual forest farmers' forest ticket trading behaviour when village collective economic organisations participate in cooperation (percentage of transactions in which foresters' forest ticket transactions were collectively supported through questionnaires or interviews)

Hypothesis 1. *Participants and strategies.* State-owned forest farms, village collective economic organisations, and individual forest farmers are chosen as the game subjects. Referring to similar studies [55,56], these three participants are assumed to have limited rationality, which means that they are not completely rational in the decision-making process but gradually adjust their strategies through trial and error and learning. The strategy choice combinations of state-owned forest farms are assumed to be {high level of inputs, low level of inputs}, with choice probabilities of g and $1-g$. The strategy choice combinations of village collective economic organisations are assumed to be {participation, non-participation}, with choice probabilities of c and $1-c$. The strategy combinations of individual forest farmers are assumed to be {high willingness to participate, low willingness to participate}, with choice probabilities of s and $1-s$, respectively.

Hypothesis 2. *Forest ticket revenue and business environment.* The income from cooperative forest management under the forest ticket system is set as R . Based on the allocation ratio of the income from forest tickets, the ratio of the value of forest tickets for state-owned forest farms, village collective economic organisations, and individual forest farmers to obtain the income from cooperative forest management is set as $a, b, d \in [0,1]$. The forest management environment is expressed by the timber price index, which is set as θ . Taking the market price of timber in the year before the implementation of the forest ticket system as the base year, $\theta > 1$ indicates that the forest management environment is favourable, and $\theta < 1$ indicates that the forest management environment is deteriorating.

Hypothesis 3. *Costs and benefits for state-owned forest farms. If the state-owned forest farm adopts a high-level forest land management strategy, it will face greater cost inputs, such as manpower, management, etc., C_1 , and, at the same time, obtain the benefits of policy implementation, B . If the state-owned forest farm operates at the input level of p under the demand for controlling the cost inputs, at this time, the cost inputs are lowered to pC_1 , and the economic and performance benefits will be proportionally reduced to apR , pB . At low levels of inputs in state forest farms, the overall benefits of forest land management are reduced, which is not conducive to maintaining good relations with surrounding communities and farmers, leading to reputational losses L_1 , and losses L_3 and nL_3 to individual foresters under different strategies.*

Hypothesis 4. *Costs and benefits for the village collective economic organisation. The village collective economic organisation's participation in forest land cooperative management generates costs C_2 in publicity and mobilisation, cooperation negotiation. and so on. If the village collective economic organisation no longer acts as a bridge, there is no cost-effectiveness generated, but there will be a loss L_2 due to the weakening of social ties with forest farmers, and additional costs C_4 and nC_4 added to the different strategy choices of individual forest farmers. In addition, at the time of forest harvesting, the village collective economic organisation will receive payment from the state-owned forest farm for the forest land use fee $M(M > C_2)$.*

Hypothesis 5. *Costs and benefits for individual forest farmers. If individual foresters have a high willingness to participate, the cost incurred in learning the relevant system and carrying out proxy agency is C_3 . On the other hand, limited by the operating characteristics of the forest land and risk-bearing ability, some foresters choose a low willingness to participate in the decision-making process, and the degree of participation is set to n . At this time, the input cost of forest farmers is nC_3 , and the gain is ndR . Meanwhile, in the process of low willingness of forest farmers to participate, there may be the phenomenon of default poaching and logging theft, which brings the risk of default to the state-owned forest farms under different strategies, which are set as G and pG .*

Hypothesis 6. *Benefits of forest ticket trading. The probability that a forest farmer participates in a forest ticket transaction is set as m ($m \in [0, 1]$), and the net gain from a forest ticket transaction is set as A . When the degree of the village collective economic organisation's role in the cooperative management of the forest land is h , the forest farmer's forest ticket transaction behaviour brings an additional prestige gain to the village collective economic organisation, hA .*

2.3.3. Model Construction

Based on the above assumptions, the game tree of cooperative forest land management under the forest ticket system is constructed, as shown in Figure 4, and the payoff matrix of the tripartite evolutionary game of state-owned forest farms, village collective economic organisations, and individual forest farmers is established, as shown in Table 2.

Table 2. Matrix of evolutionary game benefits for state-owned forest farms, village collective economic organisations and individual foresters.

Strategic Decision			Benefits		
State-Owned Forest Farms (Probability)	Village Collective Economic Organisations (Probability)	Individual Forest Farmers (Probability)	State-Owned Forest Farms	Village Collective Economic Organisations	Individual Forest Farmers
High level of inputs (g)	Participation (c)	High willingness to participate (s)	$a\theta R + B - C_1$	$b\theta R - C_2 + M + hA$	$d\theta R - C_3 + mA$
High level of inputs (g)	Participation (c)	Low willingness to participate ($1 - s$)	$a\theta R + B - C_1 - G$	$b\theta R - C_2 + M + hA$	$nd\theta R - nC_3 + mA$

Table 2. Cont.

Strategic Decision			Benefits		
State-Owned Forest Farms (Probability)	Village Collective Economic Organisations (Probability)	Individual Forest Farmers (Probability)	State-Owned Forest Farms	Village Collective Economic Organisations	Individual Forest Farmers
High level of inputs (g)	No participation ($1 - c$)	High willingness to participate (s)	$a\theta R + B - C_1$	$-L_2$	$d\theta R - C_3 + mA - C_4$
High level of inputs (g)	No participation ($1 - c$)	Low willingness to participate ($1 - s$)	$a\theta R + B - C_1 - G$	$-L_2$	$nd\theta R - nC_3 + mA - nC_4$
Low level of inputs ($1 - g$)	Participation (c)	High willingness to participate (s)	$ap\theta R + pB - pC_1 - L_1$	$b\theta R - C_2 + hA + M$	$d\theta R - C_3 + mA - L_3$
Low level of inputs ($1 - g$)	Participation (c)	Low willingness to participate ($1 - s$)	$ap\theta R + pB - pC_1 - L_1 - pG$	$b\theta R - C_2 + hA + M$	$nd\theta R - nC_3 + mA - nL_3$
Low level of inputs ($1 - g$)	No participation ($1 - c$)	High willingness to participate (s)	$ap\theta R + pB - pC_1 - L_1$	$-L_2$	$d\theta R - C_3 + mA - L_3 - C_4$
Low level of inputs ($1 - g$)	No participation ($1 - c$)	Low willingness to participate ($1 - s$)	$ap\theta R + pB - pC_1 - L_1 - pG$	$-L_2$	$nd\theta R - nC_3 + mA - nL_3 - nC_4$

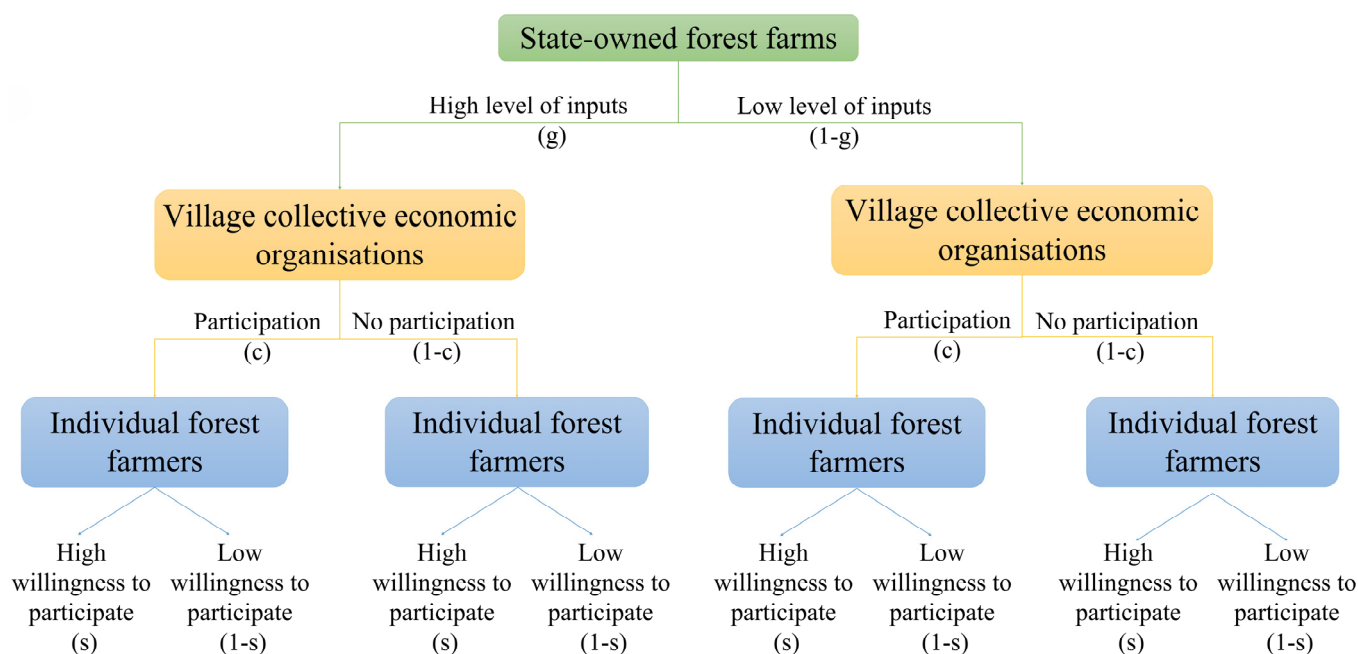


Figure 4. The game tree of cooperative forest land management under the forest ticket system.

3. Results

3.1. Results of the Game Model

In the process of evolutionary game, in order to achieve the evolution to the optimal equilibrium state, each subject is usually in the mutual game' repeated trial and error processes constantly adjust the size of the probability value of the strategy choice g , c , and s ; and this process is usually called the replication dynamic process [57]. In this paper, we solve the replication dynamic equation to explore the evolution trajectory of the strategy choices of state-owned forest farms, village collective economic organisations, and individual forest farmers in order to lay the foundation for the subsequent articles to explore the ideal equilibrium state and analyse strategy stability.

3.1.1. Analysis of Evolutionary Stabilisation Strategies of State-Owned Forestry Farms

Based on the evolutionary game payoff matrix, the expected payoff U_g from choosing a high level of inputs in the state-owned forest farms is

$$U_g = c(s(a\theta R + B - C_1) + (1 - s)(a\theta R + B - C_1 - G)) + (1 - c)(s(a\theta R + B - C_1) + (1 - s)(a\theta R + B - C_1 - G)) \quad (1)$$

The expected return U_{1-g} from choosing a high level of inputs in the state-owned forest farms is

$$U_{1-g} = c(s(ap\theta R + pB - pC_1 - L_1) + (1-s)(ap\theta R + pB - pC_1 - L_1 - pG)) + (1-c)(s(ap\theta R + pB - pC_1 - L_1) + (1-s)(ap\theta R + pB - pC_1 - L_1 - pG)). \quad (2)$$

The average expected return for the state-owned forest farms is

$$U_1 = gB + L_1g - C_1g - Gg - C_1p - L_1 - Gp - Bgp + \theta Rag + C_1gp + Ggp + Ggs + Rap + Gps - \theta Ragp - Ggps. \quad (3)$$

Reasoning from the above equation the replication dynamics equation for the state-owned forest farms can be expressed as follows:

$$F(g) = \frac{dg}{dt} = g(U_g - U_1) = -g(g-1)(B + L_1 - G - C_1 - pB + \theta Ra + C_1p + Gp + Gs - \theta Rap - Gps). \quad (4)$$

3.1.2. Analysis of Evolutionary Stabilisation Strategies of Village Collective Economic Organisations

Based on the evolutionary game payoff matrix, the expected payoff U_c for village collective economic organisations choosing to participate in cooperation is

$$U_c = g(s(b\theta R - C_2 + M + hA) + (1-s)(b\theta R - C_2 + M + hA)) + (1-g)(s(b\theta R - C_2 + M + hA) + (1-s)(b\theta R - C_2 + M + hA)). \quad (5)$$

The expected return U_{1-c} for village collective economic organisations choosing not to participate in the cooperation is

$$U_{1-c} = g(s(-L_2) + (1-s)(-L_2)) + (1-g)(s(-L_2) + (1-s)(-L_2)). \quad (6)$$

The average expected return of village collective economic organisations is

$$U_2 = L_2c - C_2c - L_2 + Mc + Ach + \theta Rbc. \quad (7)$$

Reasoning from the above equation the dynamic equation of replication of the village collective economic organisation can be expressed as follows:

$$F(c) = dc/dt = c(U_c - U_2) = -c(c-1)(L_2 - C_2 + M + Ah + \theta Rb). \quad (8)$$

3.1.3. Analysis of Evolutionary Stabilisation Strategies of Individual Forest Farmers

Based on the evolutionary game payoff matrix, the expected payoff U_s for individual foresters with high willingness to participate is

$$U_s = g(c(d\theta R - C_3 + mA) + (1-c)(d\theta R - C_3 + mA - C_4)) + (1-g)(c(d\theta R - C_3 + mA - L_3) + (1-c)(d\theta R - C_3 + mA - L_3 - C_4)). \quad (9)$$

The expected return U_{1-s} for individual foresters choosing a low willingness to participate is

$$U_{1-s} = g(c(nd\theta R - nC_3 + mA) + (1-c)(nd\theta R - nC_3 + mA - nC_4)) + (1-g)(c(nd\theta R - nC_3 + mA - nL_3) + (1-c)(nd\theta R - nC_3 + mA - nL_3 - nC_4)). \quad (10)$$

The average expected return for individual foresters is

$$U_3 = Am - C_3n - C_4n - C_3s - C_4s - L_3n - L_3s + C_4cn + C_4cs + L_3gn + C_3ns + C_4ns + L_3ns + L_3gs + \theta Rdn + \theta Rds + C_4cns - L_3gns - \theta Rdns. \quad (11)$$

The replication dynamic equation of individual foresters can be expressed by reasoning from the above formula as follows:

$$F(s) = ds/dt = s(U_s - U_3) = -s(n-1)(s-1)(C_3 + C_4 + L_3 - C_4c - L_3g - \theta Rd). \quad (12)$$

3.1.4. Stability Analysis of Equilibrium Points of Tripartite Evolutionary Game Systems

Based on the above three sets of replicated dynamic equations, by making $F(g) = 0$, $F(c) = 0$ and $F(s) = 0$, respectively, we can obtain 8 sets of pure strategy equilibria and 2 sets of mixed strategy equilibria that meet the conditions. Since the evolutionary game equilibrium must be a strict Nash equilibrium, i.e., a pure-strategy Nash equilibrium [58], for the above dynamic replicating system, only discussing the E1 (0,0,0), E2 (1,0,0), E3 (0,1,0), E4 (0,0,1), E5 (1,1,0), E6 (1,0,1), E7 (0,1,1), and E8 (1,1,1), eight levels of the stability of pure strategy equilibrium points, is necessary. According to the method proposed by Friedman [59], the stability of the equilibrium points of the system can be derived from the eigenvalue analysis of the Jacobi matrix. The Jacobi matrix J of the three-way evolutionary game system is listed first:

$$J = \begin{bmatrix} -(2g-1)(L_1 + B - G - C_1 + \theta Ra + C_1p + Gp + Gs - \theta Rap - Gps) & 0 & Gg(g-1)(p-1) \\ 0 & -(2c-1)(L_2 - C_2 + M + Ah + \theta Rb) & 0 \\ L_3s(n-1)(s-1) & C_4s(n-1)(s-1) & -(2s-1)(n-1)(C_3 + C_4 + L_3 - C_4c - L_3g - \theta Rd) \end{bmatrix}$$

Based on the Lyapunov's first method, the stability analysis is carried out for the eight sets of pure strategy equilibrium points that exist in the system. When the real part of all eigenvalues of the Jacobi matrix is negative, the equilibrium point is an evolutionarily stable strategy (ESS). When the real part of at least one eigenvalue of the Jacobi matrix is positive, the equilibrium point is unstable. When there is an eigenvalue of the Jacobi matrix with a real part of 0 and the real parts of the rest of the eigenvalues are negative, the equilibrium point is in a critical state, and it is not possible to determine its stable state by the eigenvalues [60,61]. By calculating the eigenvalues and stability of each equilibrium point through MATLAB R2022b software, three possible stable equilibrium points (0,1,0) (0,1,1) (1,1,1) can be derived. The details are shown in Table 3.

Table 3. Stability analysis of equilibrium points.

Equilibrium Point	Eigenvalues of the Jacobi Matrix			Sign of Real Part	Stability	Condition
	λ_1	λ_2	λ_3			
[0, 0, 0]	$L_2 - C_2 + M + Ah + \theta Rb$	$(n-1)(C_3 + C_4 + L_3 - \theta Rd)$	$B + L_1 - G - \frac{C_1 - pB + \theta Ra + C_1p}{Gp - \theta Rap}$	(+, *, *)	Unstable	
[1, 0, 0]	$L_2 - C_2 + M + Ah + \theta Rb$	$(n-1)(C_3 + C_4 - \theta Rd)$	$C_1 - B + G - \frac{L_1 + pB - \theta Ra - C_1p}{Gp + \theta Rap}$	(+, *, *)	Unstable	
[0, 1, 0]	$C_2 - L_2 - M - Ah - \theta Rb$	$(n-1)(C_3 + L_3 - \theta Rd)$	$B + L_1 - G - \frac{C_1 - pB + \theta Ra + C_1p}{Gp - \theta Rap}$	(-, *, *)	ESS	I
[0, 0, 1]	$L_2 - C_2 + M + Ah + \theta Rb$	$B + L_1 - \frac{C_1 - pB + Ra}{C_1p - \theta Rap}$	$-(n-1)(C_3 + C_4 + L_3 - \theta Rd)$	(+, *, *)	Unstable	
[1, 1, 0]	$(C_3 - \theta Rd)(n-1)$	$C_2 - L_2 - M - Ah - \theta Rb$	$C_1 - B + G - \frac{L_1 + pB - \theta Ra - C_1p}{Gp + \theta Rap}$	(+, -, *)	Unstable	
[1, 0, 1]	$L_2 - C_2 + M + Ah + \theta Rb$	$-(n-1)(C_3 + C_4 - \theta Rd)$	$C_1 - B - L_1 + pB - Ra - C_1p + \theta Rap$	(+, *, *)	Unstable	
[0, 1, 1]	$C_2 - L_2 - M - Ah - \theta Rb$	$-(n-1)(C_3 + L_3 - \theta Rd)$	$B + L_1 - C_1 - pB + \theta Ra + C_1p - \theta Rap$	(-, *, *)	ESS	II
[1, 1, 1]	$-(C_3 - \theta Rd)(n-1)$	$C_2 - L_2 - M - Ah - \theta Rb$	$C_1 - B - L_1 + pB - \theta Ra - C_1p + ap\theta R$	(-, -, *)	ESS	III

Note: * denotes that the sign is unknown; Condition I is $L_3 > d\theta R - C_3$, $a\theta R + B - C_1 - G < (ap\theta R + pB - pC_1 - pG) - L_1$; Condition II is $a\theta R + B - C_1 < (ap\theta R + pB - pC_1) - L_1$, $L_3 < d\theta R - C_3$; and Condition III is $(ap\theta R - pC_1 + pB) - L_1 < a\theta R + B - C_1$.

According to the eigenvalues of the Jacobi matrix of the game model, the possible equilibrium states of the tripartite game subjects of state-owned forest farms, village collective economic organizations, and individual forest farmers are further explored as follows:

Case 1—When $(a\theta R - C_1) - G + B < (ap\theta R - pC_1) - pG + pB - L_1$, $0 > d\theta R - C_3 - L_3$ (0,1,0) is the evolutionary stabilization point. At this point, the forest management environment is in a low period, the net benefit of high level of input from state-owned forest farms is less than the low level of input, the net benefit of high willingness of individual forest farmers to participate in cooperative forest management is negative, and the three parties are stabilized in the {low level of input, participation, low willingness to participate} strategy. Due to the downturn in the forest market environment, the low level of returns from forest management, and the weak trust relationship between state-owned forest farms and individual forest farmers due to their lack of understanding of each other, inefficient management and default logging easily occur in cooperative management, with both parties experiencing lower returns and reduced levels of commitment and willingness. At this time, the village collective economic organisation uses the advantages of grass-roots economic organisations to gather foresters and forest land resources while assuming some of the risks of cooperation, and participates in the cooperative management of forest land in the form of a collective, which makes it possible to obtain economic and social benefits that are significantly higher than the costs. However, as time progresses, the long-term inefficient operation of state-owned forest farms and the defaults brought about by the low willingness of individual foresters to participate have made the costs and risks of village collective economic organizations' participation in cooperative management rise, which are not conducive to the long-term effective operation of cooperative management.

Case 2—When $a\theta R + B - C_1 < (ap\theta R + pB - pC_1) - L_1$, $0 < d\theta R - C_3 - L_3$ (0,1,1) is the evolutionary stability point. At this point, the forest management environment remains stable with the early stage of cooperation, the net benefit of high-level input from state-owned forest farms is smaller than that of low-level input, the net benefit of the forest ticket when individual foresters have a high willingness to participate in cooperative forest land management is positive, and the three parties are stabilized in the {lower level of input, participation, and high willingness to participate} strategy. In this operating environment, given the professionalism of state-owned forest land management, the level and benefits of its forest land management are generally higher than those of decentralized management. In order to achieve the goal of maximizing the benefits of forest land, village collective economic organizations, and individual foresters are more inclined to join the cooperative operation in the form of forest land shares to achieve double growth in economic and social benefits. In the short term, the results of forest land management have improved significantly due to the provision of sufficient forest land resources to support cooperative management. However, in the medium and long terms, due to the low level of input behaviour of state-owned forest farms, the cooperative management of forest land has not realized the intensive cultivation, the phenomenon of waste of forest land resources has increased, and the operating efficiency has diminished, which directly affect the income of cooperative management. At the same time, the negative cooperative behaviour of the state-owned forest farms will also lead to a decline in the trust of the surrounding communities and farmers, resulting in a deterioration of the relationship.

Case 3—When $(ap\theta R - pC_1 + pB) - L_1 < a\theta R + B - C_1$, (1,1,1) is the evolutionary stabilization point. At this point, in the forest management environment continues to evolve, the net benefit of high level of input from state-owned forest farms is greater than that of low level of input, and the tripartite is stabilized at the {high level of input, participation, high willingness to participate} strategy. In that operating environment, state-owned forest farms give full play to their advantages in forest land management. Through the leading talent and technology advantages, they attract more village collective economic organizations and forest farmers to participate in the cooperative management so as to achieve the maximization of operating efficiency and the goal of “double growth”.

The village collective economic organization fully coordinates the interests between the state-owned forest farms and individual forest farmers. Through cooperation negotiations with the state-owned forest farms, the organization seeks to maximize the collective benefits of cooperation and achieve the village collective financial income. Through training and education for forest farmers, it strengthens their understanding and awareness of cooperative management, deepens social ties with forest farmers, and realizes increased social prestige. Individual forest farmers are willing to invest time and level through the state-owned forest farms and village collective economic organizations to have a deeper understanding of cooperative management, fully invest in forest resources, performance and trustworthiness, and maximize the return on the value of the forest ticket. This situation is a more ideal state and has an important role in promoting the effective operation of the forest land cooperative management of the forest ticket system.

In addition to the above three cases, when Conditions I and III are satisfied at the same time, there exist two evolutionary stable points, $(0,1,0)$ and $(1,1,1)$, which present a bistable situation. At this point, $0 > d\theta R - C_3 - L_3$ and $(1-p)(C_1 - a\theta R - B) < L_1 < (1-p)(C_1 - a\theta R + G - B)$, the cooperation takes two forms. In the first form, the three parties stabilise in the {low level of input, participation, low willingness to participate} strategy. In terms of the development history of China's forestry industry, the ownership of collective forest land in the south has long belonged to the collective, and the participation of state-owned forest farms and individual foresters in the management of collective forest land is low. Therefore, due to historical experience as well as path dependence, village collectives tend to take the initiative to promote cooperation as a leader, while state-owned forest farms and individual forest farmers present a negative strategy towards the possibility of participating in cooperation. The three parties in the second form stabilise at the {high level of input, participation, and high willingness to participate} strategy. As the reform of China's collective forest rights system continues to advance, the three rights of forest land are divided and the restrictions on forest land transfer are constantly being liberalised. Cooperative forest management among state-owned forest farms, village collective economic organisations, and individual forest farmers with the goal of increasing the total utility of forest land management has received policy support. In cooperative decision-making, the three parties are more inclined to make positive decisions. The establishment of the cooperative forest land management model under the forest ticket system aims to encourage the intensification of forest land resources and the enhancement of management inputs through the transfer of forest rights so as to transform the ecological value of forests into economic value. However, the cooperation pattern represented by the stability point $(0,1,0)$ is not conducive to the effective operation of the system in the long term and is not in line with the original intention of the system. Therefore, it is important to promote the stable evolution of $(1,1,1)$ to achieve the win-win situation of the three parties in the decision-making of the three parties to participate in the cooperative management of forest land, which is of great significance to promote the effective operation of the system of forest stamps in the long term.

3.2. Simulation Analysis Results

3.2.1. Effects of Changes in Initial Probability

Through the above analyses, combined with the real needs and implications of the pilot cases, the most ideal equilibrium state $(1,1,1)$ is analysed to explore the influence of the key parameters on the evolution of the behavioural strategies of the game subjects. Combined with the pilot case of the forest ticket, the management method of the forest ticket

system and the requirements of the evolutionary game method on the specific parameters, the specific parameters are initially assigned values. The forest management environment is assigned $\theta = 1$ based on the unchanged base year. The initial value of the forest ticket and the share of equity allocation of the forest ticket in Yuzhu Village are assigned $R = 100$, $a = 0.51$, $b = 0.147$, and $d = 0.343$. According to the management method of the forest ticket system, the village collective economic organisation receives the state-owned forest land use fee $M = 10$. Adopting the state-owned forest farm forest ticket pocket repurchase interest rate is assigned $A = 35.329$. Referring to the research of Zhang Xiumei and other [46] scholars and the actual production situation, the values $C_1 = 20$, $C_2 = 8$, $C_3 = 5$, and $B = 5$ are assigned, and the individual foresters need to bear additional costs $C_4 = 3$ when the village collective economic organisation is not involved. Referring to the finite rational production decision under information asymmetry, the initial degree of participation is assigned as $p = 0.5$ and $n = 0.5$ when state-owned forest farms and individual foresters take negative participation decision, the initial probability is assigned as $m = 0.5$ for individual foresters to participate in the transaction of forest tickets, and the degree of the role of village collective economic organisations in the co-operative management of forest land is assigned $h = 0.5$. The values $L_1 = 10$, $L_2 = 3$, $L_3 = 5$, and $G = 10$ are assigned with reference to the parameter design and game equilibrium point requirements.

The evolutionary trajectory of the equilibrium strategy at point E8 (1,1,1) in Table 4, i.e., when all three adopt an active strategy, is simulated by MATLAB R2022b. The horizontal axis denotes time t , and the vertical axis denotes the probability of positive behavioural decisions taken by state-owned forest farms, village collective economic organisations, and individual foresters. Different initial wills are assigned to the three subjects, as shown in Figure 5. All three participating subjects with a different initial willingness stabilise over time to actively participate in the cooperative management strategy, where the decision-making speed is faster for village collective economic organisations than for state-owned forest farms than for forest farmers. The higher the initial willingness of each subject, the faster their convergence.

This fully demonstrates that in the cooperative forest land management under the forest ticket system, village collective economic organisations give full play to their role as a bridge and intermediary [62]. In cooperation, they use their own organisational advantages to unite and organise the farmers, gather forest land resources, provide a large amount of forest land capital for the forest land operation of the state-owned forest farms, improve the potential income, and attract the state-owned forest farms to invest in the forest land operation at a high level. State-owned forest farms, with the strong support of village collective economic organisations, make full use of the advantages of technology and experience to expand the scale and benefits of forest land operation and maximise the benefits. Under the propaganda and training of state-owned forest farms and village collective economic organisations, individual forest farmers break the information barriers, fully understand the benefits and risks of forest land cooperation, and are more willing to actively invest in forest land cooperation when integrating their own visions of increasing income. The initial willingness to participate in the cooperative forest management model under the forest ticket system fluctuates among the participating subjects due to the influence of their knowledge and recognition of the relevant policy system and the cooperative model. The higher the degree of recognition of the cooperative forest management model, the higher the initial willingness to participate, and the faster the speed of finalising the cooperation. However, the changes and adjustments of the decision variables of each subject will affect the evolution path of the system's equilibrium strategy, and so it is necessary to further analyse the mechanism of the influence of important decision variables on the system's evolution equilibrium in the replication dynamic system.

Table 4. Similarities and differences between SOTA and this research work.

Dimension	SOTA Methods	The Research in This Paper
Theoretical Assumptions	Based on finite rationality but mostly assumes subject homogeneity.	Based on limited rationality, emphasising subject heterogeneity (differences in cognitive level and ability to acquire information) and reflecting the incompleteness of the exploration stage.
Modelling Approach	Complex networks, stochastic evolution, fractional order models.	Based on the framework of evolutionary game analysis, combining the reward and punishment mechanism with marginal strategy adjustment in the policy case.
Application Scenarios	Mature environmental policy or cross-disciplinary cooperation.	Focusing on the emerging forest ticket system, analysing the equilibrium of interests and evolution paths of multiple subjects at the early stage of the policy, emphasising long-term dynamics and policy adaptability.
Simulation Technology	MATLAB-based multi-subject numerical simulation.	MATLAB-based multi-subject numerical simulation with simulation values based on real-life cases.
Methodological Innovations	Introduction of complex networks or stochastic perturbations to enhance model realism.	Combining the real cases and policy background, exploring the dynamic trial-and-error characteristics of the stage, and revealing the optimisation mechanism of the subject's strategy.

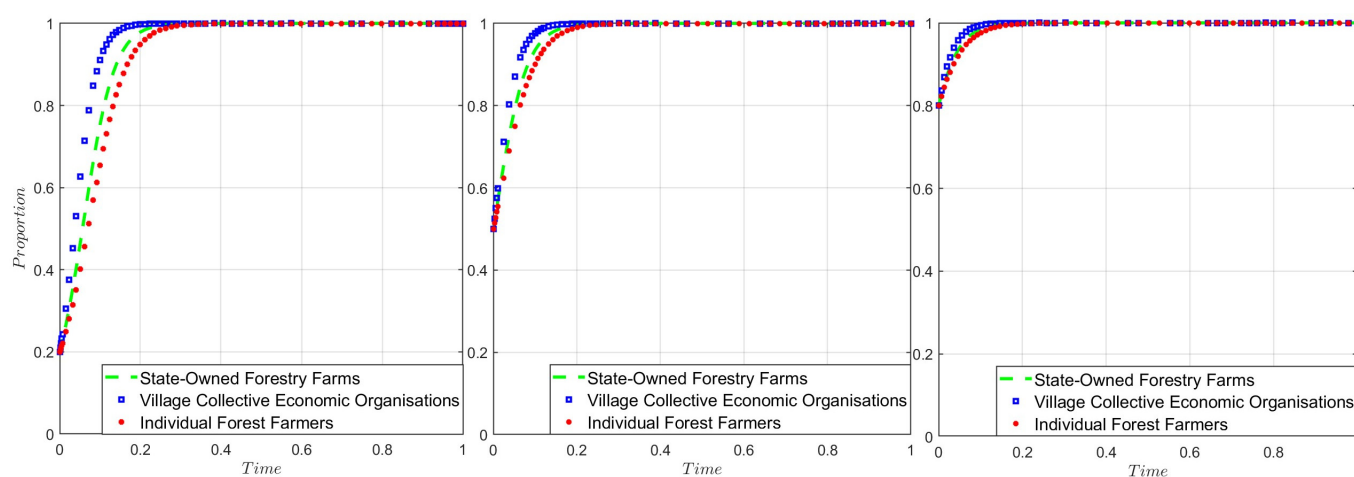


Figure 5. Effects of changes in the initial probability.

3.2.2. Simulation Results for the Forest Business Environment

During one or even more logging rotations, the forest management environment and the vigour of the timber sales market have direct impacts on the expected returns of cooperative forest land management under the forest ticket system of each subject. Therefore, in order to analyse the evolutionary impacts of changes in the forest management environment on the behavioural strategy choices of different subjects, the simulation analysis was carried out by assigning $\theta = 0.5, 1$, and 1.5 according to the low, medium, and high levels, respectively, while keeping other parameters unchanged, with reference to the methods of Qiao et al. [63]. The details are shown in Figure 6. In the process of system evolution equilibrium, as the forest management environment develops in a positive direction and market vitality increases, the probability of state-owned forest farms, village collective economic organisations, and forest farmers adopting positive cooperation strategies all increase to different degrees, and the time to reach equilibrium decreases significantly. On the contrary, when the forest management environment develops in an unfavourable direction and the timber market is depressed, the probability of all three choosing positive cooperation strategies decreases, and the time to reach equilibrium is significantly lengthened. Among them, as the main operators of forest land cooperation and providers of forest land production factors under the forest ticket system, state-owned forest farms and individual forest farmers are much more sensitive to the forest management environment than village collective economic organisations.

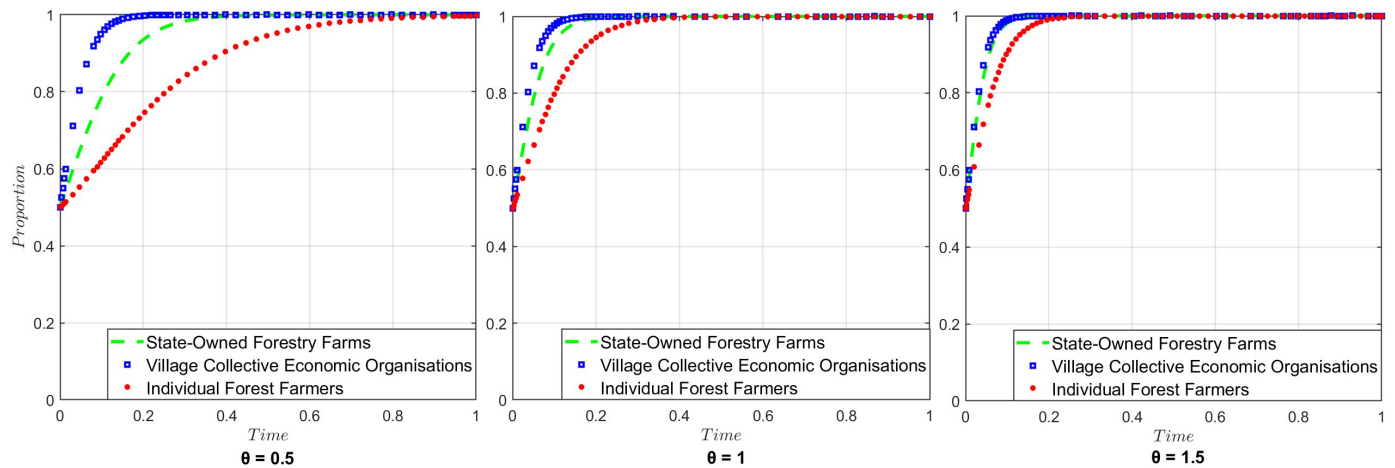


Figure 6. Impacts of the forest business environment.

3.2.3. Simulation Results for the Level of Inputs to State-Owned Forest Farms

In order to clarify the influence of the degree of inputs of state-owned forest farms on the evolution results of the system, simulation analyses were carried out by assigning $p = 0.2, 0.4, 0.6$, and 0.8 , respectively, under the premise of keeping the other parameters unchanged. The details are shown in Figure 7. Influenced by the law of diminishing marginal benefits, the incremental benefits brought by each additional unit of input of means of production in state-owned forest farms are decreasing, while there is an upper limit to the total maximisation of the benefits of forest land management. In the process of system evolution to the stable point, for the state-owned forest farms with increasing degrees of business inputs, the speed of convergence of their strategies to choose high levels of inputs becomes slower and slower, and the time to reach equilibrium increases. The behavioural strategy selection paths of village collective economic organisations and individual forest farmers, on the other hand, do not change significantly.

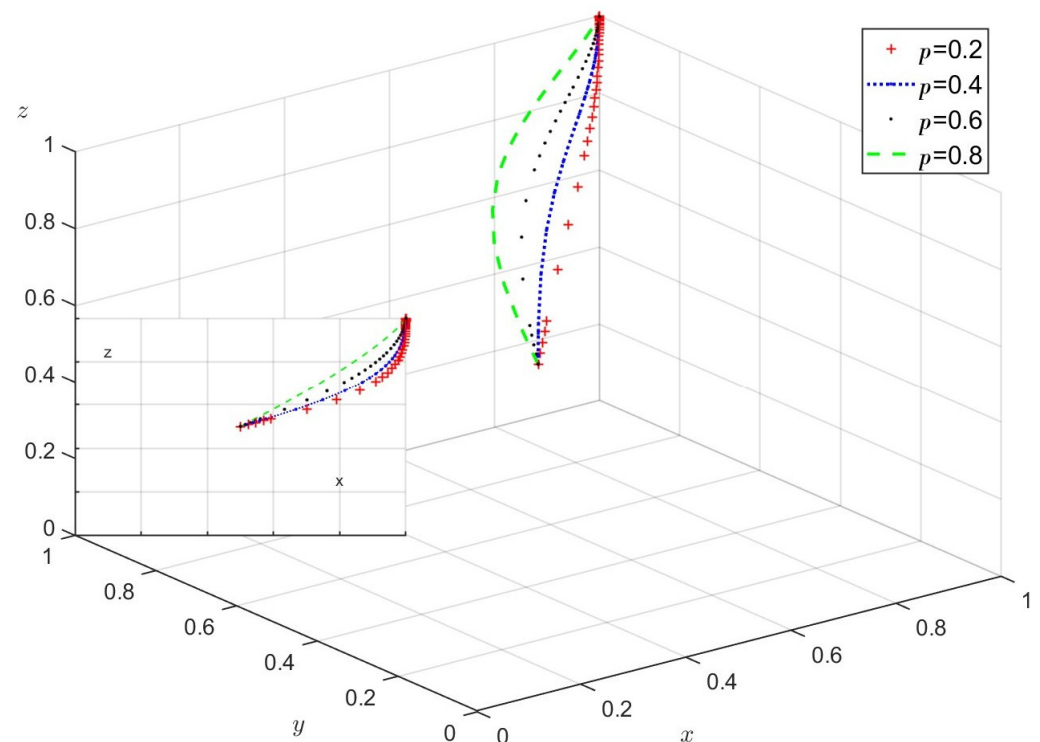


Figure 7. Impacts of the level of inputs from state-owned forests.

3.2.4. Simulation Results for Village Collective Economic Organisations' Participation Costs

In order to clarify the impact of the costs incurred by the village collective economic organisations participating in the cooperative management of forest land on the system evolution results, under the premise of keeping the other parameters unchanged, the simulation analysis was assigned $C2 = 8, 23$, and 38 , respectively. In the equilibrium process of system evolution, as the cost of village collective economic organisations to participate in forest land cooperation increases, the convergence speed of their final choice of participation in the cooperation strategy will be slower and slower. At this time, the strategy choice of state-owned forest farms and individual forest farmers is relatively stable, specifically as shown in Figure 8.

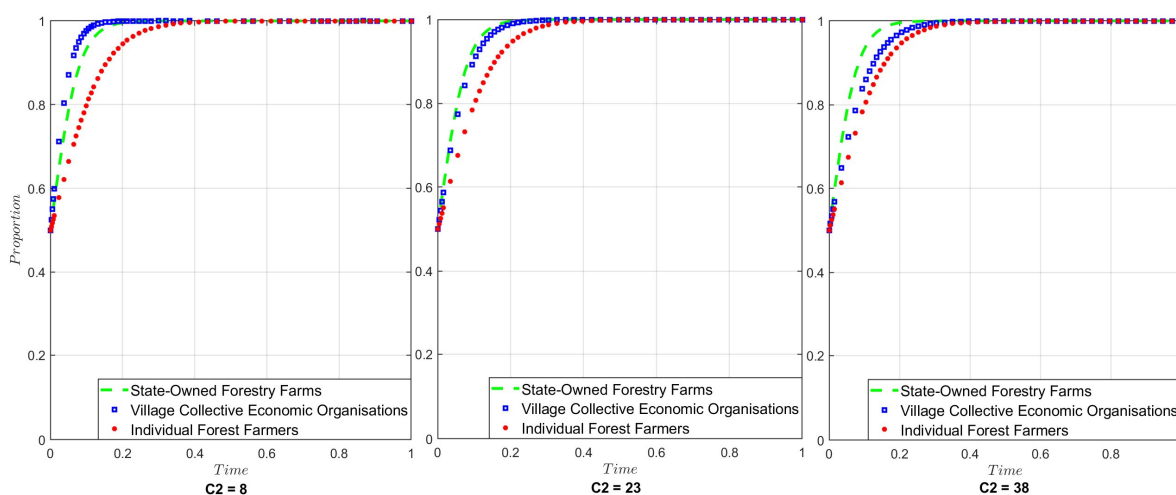


Figure 8. Impacts of the costs incurred by village collective economic organisations participating in co-operation.

3.2.5. Simulation Results for the Degree of Participation of Individual Foresters and the Impact of Forest Ticket Transaction Benefits

In order to clarify the influence of the degree of participation of individual foresters on the evolution results of the system, under the premise of keeping other parameters unchanged, $n = 0.2, 0.4, 0.6$, and 0.8 were assigned to carry out simulation analyses, respectively. Restricted by the characteristics of smallholder farmers, individual foresters have high economic vulnerability and weak risk tolerance, and are more sensitive to possible operational fluctuations in forest land cooperation [64]. In the process of system evolution to the stable point, as the degree of participation continues to increase, individual forest farmers ultimately choose a high willingness to participate in the cooperation strategy, the convergence speed will be slower and slower, and the time to reach equilibrium increases, while the state-owned forest farms' and village collective economic organisations' behaviour of the strategy selection path will not change significantly, specifically as shown in Figure 9.

In the cooperative management of forest land, the transaction income of forest ticket obtained by forest farmers through participation in the cooperation is also an important factor affecting their decision-making. In order to clarify the influence of the expected income from the forest ticket transaction on the evolution results of the system, under the premise of keeping the other parameters unchanged, the simulation analysis is assigned to $A = 20, 35$, and 50 , respectively, as shown in Figure 10. In the forest ticket system, the main goal of forest farmers to participate in the cooperation is to improve their overall income situation. During the evolution of the system to the stable point, as the expected

returns from the forest ticket transaction continue to improve, the degree to which the economic interests of individual foresters are satisfied increases. Therefore, individual farmers are more willing to participate in cooperation to convert forest resources into forest ticket income, and the convergence speed of the strategy of a high willingness to participate in the cooperation will be faster and faster.

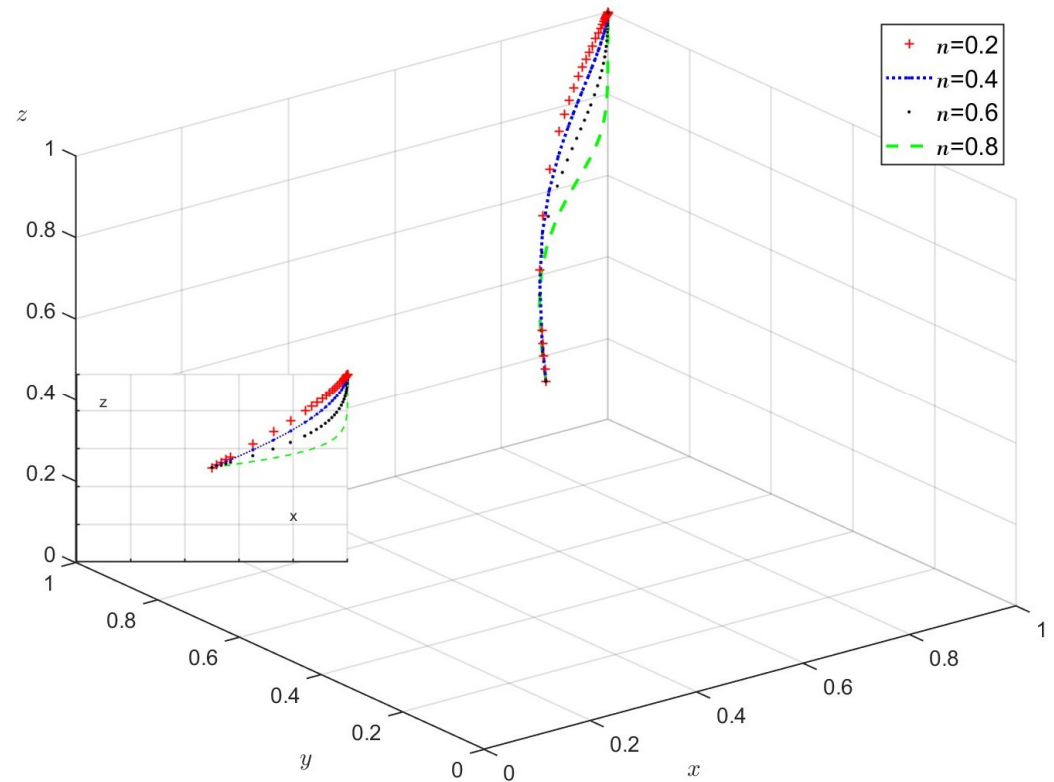


Figure 9. Impact of the level of participation of individual foresters.

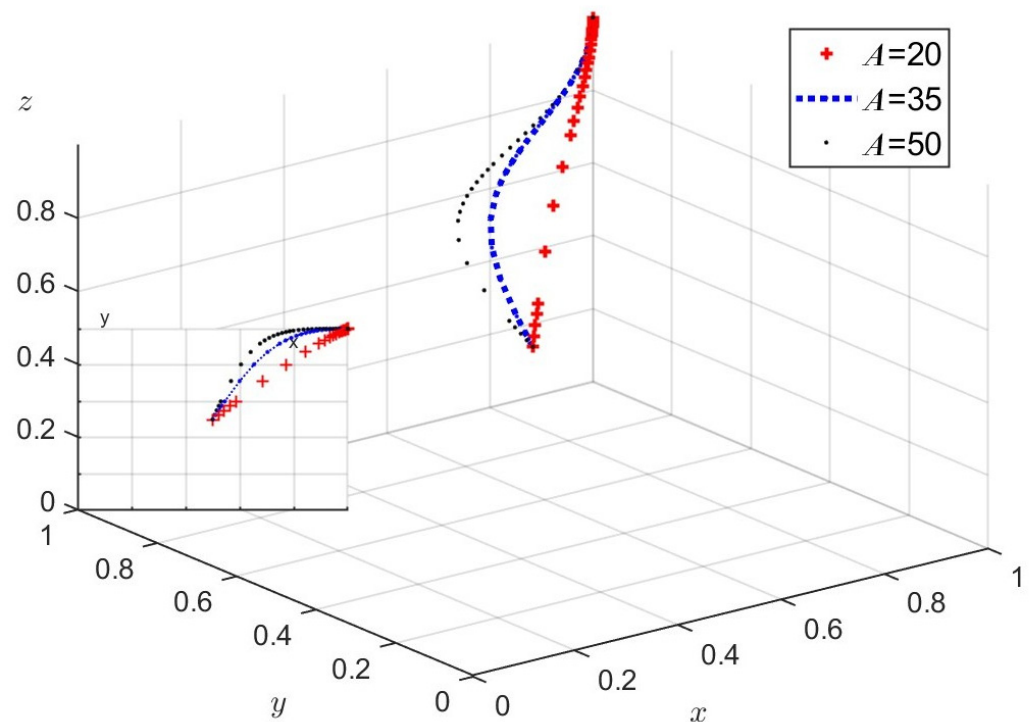


Figure 10. Impact of proceeds from forest ticket trading.

4. Discussion

Since the implementation of the forest ticket system, Sanming City, Fujian Province, has promoted the forest ticket mode of cooperation between state-owned forestry enterprises and institutions, local village collective economic organisations, and individual foresters, and has initiated a pilot scheme in 411 villages in the city, with a cooperation area of 304,000 mu and a total of 768 million yuan in forest tickets issued. The pilot villages can increase the village financial income by more than 50,000 RMB per year by collecting forest land use fees and so on [20]. However, past studies have shown that the development of forest land cooperative management under the forest ticket system still faces a series of challenges, such as breach of contract and deforestation behaviours, conflicts of interest, etc. [65,66]. How to coordinate the good relationship between stakeholders is the key influencing factor to promote the participation of multiple subjects in forest land cooperative management under the forest ticket system and to promote the long-term operation of the forest ticket system.

This paper analyses the strategic behaviour of the three-party game of ‘state-owned forest farms, village collective economic organisations, and individual forest farmers’ in the cooperative forest management under the forest ticket system, taking the practice of the forest ticket system in Sanming City, Fujian Province, China, as an example. The results of the game model show that there is a stable equilibrium strategy in the three-party game, which can achieve a win–win situation for all three parties, and, at the same time, promote the sustainable management of forests, the economic growth of forestry, the unbundling of human–land relations, and an increase in the income of forest farmers. In the three-party equilibrium game, the behavioural decisions of state-owned forest farms are pivotal to the quality and effectiveness of forest land cooperative management under the forest ticket system, and their low level of input behaviour will directly affect the behavioural willingness and cooperative effects of other subjects, similar to the results of previous studies [67]. The strategy choice of village collective economic organisations focuses more on their own benefits [62], i.e., they tend to take the initiative to participate in cooperative management when they can obtain higher benefits. Individual forest farmers have the basic production factors of forest land cooperative management, and the high willingness of individual forest farmers to participate in cooperation is an indispensable prerequisite for promoting forest land cooperative management under the forest ticket system [67]. Therefore, how to promote the state-owned forest farms, village collective economic organisations, and individual forest farmers to give full play to their respective functional advantages and efficient synergistic cooperation has an important impact on enhancing the level of cooperative forest land management.

In addition, this paper provides an in-depth understanding of the actual situation through interviews and surveys, and numerical simulations are carried out by combining the case data of the Fujian Provincial Forestry Bureau, news reports, and the related literature. This provides a decision-making reference for the long-term development of the forest ticket system. We find that there are several key issues that need to be focused on to promote the effective operation of the forest ticket system. First, fluctuations in the forest business environment can significantly affect the returns of the whole forest ticket system, which in turn affects the decision-making choices of both state-owned forest farms and individual foresters. The former, as an important player in direct contact with the forest trading market, as well as the main operator of the forest land, is more sensitive to changes in returns due to market fluctuations [68]. The latter, due to a weak risk-taking ability, have insufficient decision stability in the face of changes in the operating environment [64]. Secondly, state-owned forest farms are of a public welfare nature, and in the process of forest ecological construction, they are often faced with the dual pressure of insufficient

funds and difficulty in expanding the scale of operation [34]. In view of its ecological and economic dual-task needs, increasing the net income from forest land management has become a direct driving force for state-owned forest farms to enhance the degree of input. Third, as an intermediary role, village collective economic organisations can mobilise forest farmers and have a facilitating effect on the achievement of forest ticket cooperation [62]. However, due to its relatively fixed source of income, the range of economic returns that can be fluctuated is very limited. This leads to the village collective economic organisation being very sensitive to changes in cost inputs. Fourth, economic returns are the main factor influencing forest farmers and their families to make major decisions [69]. Compared with the self-owned land management, the investment of land as capital in forest land cooperative management brings more significant benefits to individual forest farmers [13]. However, limited by the characteristics of smallholder farmers, the risk aversion preference of forest farmers and information asymmetry hinder the willingness of forest farmers to participate.

In order to better promote the long-term effective operation of the forest ticket system and provide new perspectives for the sustainable management of forest land, based on the analysis in this paper, the following optimisation strategies can be adopted from the perspective of stakeholders. First, the cooperative operation of forest land should be promoted to reduce costs and improve operational efficiency. The government should continue to improve the financial allocation and subsidy mechanism to mitigate the participation costs and operational risks of each participating subject. State-owned forest farms should implement refined forest management and use scientific forest management tools combined with big data analysis to achieve cost reduction and efficiency in forest management. Second, the construction of a risk guarantee mechanism should be improved to reduce the risk of forest ticket cooperation and transaction. The government, as a supervisory and guiding unit, should discuss the formulation or revision of relevant laws and regulations to promote the construction of the forest ticket transaction mechanism and reduce the risk of forest farmers' forest ticket transactions. The market should establish an early warning and emergency response mechanism for the risk of the forest ticket system to prevent the risk of economic market changes and natural disaster risks. It should also introduce a third-party supervisory organisation to provide service support in every aspect of forest ticket transactions. Finally, actively carrying out publicity and mobilisation to break down information barriers and promoting the participation rate are needed. Policy propaganda should be strengthened, a digital platform for forest ticket information services should be built, and typical case visits should be carried out to improve public awareness. Village collective economic organisations should strengthen theoretical learning, constantly cultivate risk awareness and risk-resistant ability, and launch extensive publicity for forest farmers to break their thinking limitations. Individual foresters should actively participate in various types of training at all levels, such as science forums and knowledge visits to the countryside, so as to enhance their awareness of and sensitivity to the forest ticket system, and to improve their sensitivity in identifying and grasping opportunities.

5. Conclusions and Recommendations

5.1. Conclusions

Based on promoting the transformation of forest resources into economic capital and exploring the path of balanced economic and ecological development under the global sustainability goal, this paper conducts further research on the construction of a cooperative model for China's forest ticket system. By constructing a three-party evolutionary game model of 'state-owned forest farms–village collective economic organisations–individual forest farmers' from the perspective of synergy of multiple subjects and analysing the

stability of the model using the Li Yapuno first method, combined with the simulation analysis of the pilot areas of the forest ticket reform, we have systematically explored the effects of different influencing factors on the stable state of the evolutionary game was explored. The study found that

- (1) The long-term sustainable operation of the forest ticket system requires the collaborative participation of multiple stakeholders, and the three stakeholder groups will influence each other's strategy choices in the game process. Under certain conditions, the three subjects can achieve three stable equilibrium strategies—(0,1,0), (0,1,1) and (1,1,1). Among them, (1,1,1), i.e., a high level of input from state-owned forest farms, participation of village collective economic organisations, and high willingness of individual foresters to participate, is the most realistic and ideal scenario.
- (2) The degree of recognition and participation of each subject in the forest ticket system helps to promote the three parties to reach the optimal equilibrium strategy. When the degree of information asymmetry is significantly reduced, the flow of information between participating subjects is more transparent. At this time, the flow of information among the subjects is smoother and the degree of access to information is similar. State-owned forest farms can formulate a good management strategy and increase the investment in forestry by playing their professional advantages. Village collective economic organizations can actively play the role of a link to build a good bridge of communication. Individual foresters have a high level of trust in the cooperative partners and model, and the risk of default on deforestation is reduced. Therefore, breaking down information barriers can effectively reach a win-win situation in multi-party collaboration and promote the orderly operation of the forest ticket system.
- (3) The development of the forest management environment and market environment to good or bad will significantly affect the behavioural decision-making and the formation of cooperation equilibrium of each subject. When the market operating environment is expected to develop favourably, state-owned forest farms and individual foresters are more inclined to participate in cooperation with high input and a high willingness to promote the rapid achievement of cooperation. Conversely, there are more challenges to achieving cooperation. Therefore, optimising the cooperation mechanism, improving the risk protection mechanism, and enhancing the ability to withstand market fluctuations are crucial to promote the sustainable development of forest land cooperation.
- (4) Changes in the costs and benefits of cooperative forest management are closely related to the decision-making choices of each subject. State-owned forest farms, as the leading party of forest land cooperation, are particularly sensitive to changes in the cost-benefit ratio. When the marginal return brought by the input level decreases, the state-owned forest farms tend to reduce inputs and lower input levels, thus falling into the vicious circle of 'lower input levels-lower marginal returns-lower operating incentives-lower input levels'. Therefore, improving management efficiency and controlling the ratio of income to cost is an important means to promote forest land cooperation.
- (5) The tradable and pledge characteristics of the forest ticket provide individual foresters with a way to increase their income beyond the income from forest land, effectively motivating individual foresters to actively participate in cooperative forest land management. The higher the degree of participation and willingness of individual foresters, the greater the possibility of obtaining additional economic benefits through the transfer of forest tickets and financial pledges, thus improving the overall level of returns.

5.2. Research Limitations and Prospects

Overall, although this study has made progress in exploring the decision-making game of multiple subjects' participation in forest land cooperation under the forest ticket system, promoting the effective operation of the forest ticket system, and providing policy recommendations for the optimisation path of the development of the forest ticket system, there are several limitations that deserve to be acknowledged:

- (1) In today's academic research field, the methodology of evolutionary game models is undergoing rapid changes and development. Cutting-edge research methods such as stochastic evolutionary models and fractional order models are gradually and widely introduced into the analysis of evolutionary games, injecting new vitality and ideas into the research in this field. However, we must honestly admit that there is still a gap between the methodologies used in our research work and the most advanced and exquisite research methods in the international arena, which requires us to make continuous efforts to catch up and improve. We have collated the similarities and differences between SOTA and this research work below.
- (2) Based on the information and data from visits and interviews with representatives of forestry government departments, forestry enterprises and institutions, village collective economic organisations, and forest farmers in the Sanming region of Fujian, the generalisability of the research results may be limited.
- (3) Because the forest ticket system is still a new thing in China's forestry sustainable management policy system, the information available on the revenue of forest land cooperation is limited, and the current study discusses the theoretical level and the institutional level, but the analysis of the micro level needs to be improved.
- (4) According to the existing policies and documents of the forest ticket system, the government departments only play the role of encouraging and supervising, and do not actually participate in the specific cooperative management and the process of forest ticket distribution. Therefore, this study focused on state-owned forest farms, village collective economic organisations, and forest farmers. However, with the continuous improvement of the forest ticket system, it is not yet known how the role of government departments will change in the future. Future research should follow the adjustment and implementation of the policy, and explore the impact of the forest ticket system on the sustainable management of China's forestry industry in more depth so as to enrich our understanding of this innovative forestry management system.

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